How-To Library

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# Importing and Exporting Data Assets Between Microsoft Azure SQL Database and Enterprise Data Preparation

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# Abstract

You can import data assets from Microsoft Azure SQL Database to your data lake, and can export data assets from your data lake to Azure SQL Database. Review this article to prepare Enterprise Data Preparation to import and export data assets.

# Supported Versions

- Enterprise Data Preparation 10.2 x
- Enterprise Data Preparation 10.4 x

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# Overview

You can import data from Azure SQL Database tables into your data lake. You can also export data from your data lake to Azure SQL Database.

After you import a data asset from Azure SQL Database, you can add the imported data to a project in Enterprise Data Preparation as a worksheet. After you prepare an asset and export the data to Azure SQL Database, you can perform further operations on the data in Azure SQL Database.

**Note:** You must export assets to the `default` schema in the Azure SQL Database target. You cannot export an asset to a different schema.

To prepare to import and export data assets between Azure SQL Database and Enterprise Data Preparation, complete the following steps:

1. Download the Microsoft JDBC Driver for SQL Server, and then copy the driver to the Data Integration Service machine associated with Enterprise Data Preparation.
2. Create one or more JDBC connections to import and export data assets. You might create different connections for import and export operations.
3. Add the DoNotUseOwnerNameForSqoop custom property to the Enterprise Data Preparation Service.

# Download the Microsoft JDBC Driver for SQL Server

Download the Microsoft JDBC Driver for SQL Server, and then copy the driver to the Data Integration Service machine associated with Enterprise Data Preparation.

When you import or export data, the Data Integration Service uses Sqoop to connect to the cluster, and then executes a Sqoop job on the cluster to import or export the data. Sqoop uses the driver to connect to the Azure SQL Database target when the import or export job runs on the cluster.

Download the Microsoft JDBC Driver for SQL Server .jar file that corresponds to your preferred Java Runtime Environment (JRE) settings. You can download the .jar file from the following link:

https://www.microsoft.com/en-us/download/search.aspx?q=jdbc

Copy the .jar file to the following directory on the machine where the Data Integration Service runs:

```
<Informatica installation directory>/externaljdbcjars
```

## Create JDBC Connections to Access Azure SQL Database

Create one or more JDBC connections to import and export data assets. You might create different connections for import and export operations.

You include the arguments Sqoop uses to connect to the Azure SQL Database target in each JDBC connection.

1. In the Administrator tool, click the **Manage** tab > **Connections** view.
2. In the Domain Navigator, click **Actions > New > Connection**.

   The New Connection dialog box appears.
3. In the New Connection dialog box, select **Database** > **JDBC**.
4. Click **OK**.

   The **New Connection** wizard appears.
5. Enter the JDBC connection properties required to connect to the Azure SQL Database target.

   The following table describes the connection properties you enter:

| Property | Description |
|---|---|
| Name | Name of the connection. The name is not case sensitive, and must be unique within the domain. The name cannot exceed 128 characters, contain spaces, or contain the following special characters:<br>`~ ` ! $ % ^ & * ( ) - + = { [ } ] \| \ : ; " ' < , > . ? /` |
| ID | String that the Informatica services use to identify the connection. The string is not case sensitive. The string must be 255 characters or less, and must be unique in the domain. Default value is the connection name. |
| Description | Description of the connection. The description cannot exceed 765 characters. |
| User Name | Database user name. |
| Password | Password for the database user name. |
| JDBC Driver Class Name | Name of the JDBC driver class packaged with Informatica. Enterprise Data Preparation uses the driver to connect to the Azure SQL Database target. Enter the following value:<br>`com.informatica.jdbc.sqlserver.SQLServerDriver` |
| Connection String | Connection string to connect to the target. Use the following format for the connection string:<br>`jdbc:informatica:sqlserver://<database host name>:<database port>;databaseName=<database name>;EncryptionMethod=SSL;HostNameInCertificate=*.<secure database host name>;validateServerCertificate=false;`<br>The following example shows a sample connection string:<br>`jdbc:informatica:sqlserver://mysqldb.database.windows.net:1433;databaseName=edlsqldb;EncryptionMethod=SSL;HostNameInCertificate=*.database.windows.net;ValidateServerCertificate=false;` |

| Property | Description |
|---|---|
| Environment SQL | Optional. SQL commands to set the database environment when you connect to the target. The Data Integration Service executes the connection environment SQL each time it connects to the target. |
| Transaction SQL | Optional. SQL commands to set the database environment when you connect to the target. The Data Integration Service executes the SQL at the beginning of each transaction. |
| SQL Identifier Character | Select " " from the drop down list to connect to the target from a Hortonworks HDP cluster or from a Cloudera CDH cluster. |
| Support Mixed-case Identifiers | Select **Yes** from the drop down list. |
| Use Sqoop Connector | Select **Sqoop v1.x** to enable Sqoop connectivity. |
| Sqoop Arguments | Sqoop arguments required to connect to the target.<br>Specify the connection string Sqoop uses to connect to the target in the `--connect` argument. Specify the class name of the Microsoft JDBC Driver for SQL Server driver Sqoop uses in the `--driver` argument.<br>Use the following format for the argument string:<br>`--connect jdbc:sqlserver://jdbc:informatica:sqlserver://<database host name>:<database port>;databaseName=<database name>;;encrypt=<true\|false>;trustServerCertificate=<true\|false>;hostNameInCertificate=*.<secure_database_host_name>;loginTimeout=<timeout in seconds> --driver <JDBC driver name>`<br>The following example shows a sample argument string:<br>`--connect jdbc:sqlserver://edlsqldb.database.windows.net:1433;database=mysqldb;encrypt=true;trustServerCertificate=false;hostNameInCertificate=*.database.windows.net;loginTimeout=30 --driver com.microsoft.sqlserver.jdbc.SQLServerDriver` |

6. When you finish entering the connection properties, click **Test Connection** to test the connection.
7. Click **Finish**.

# Configure the DoNotUseOwnerNameForSqoop Custom Option

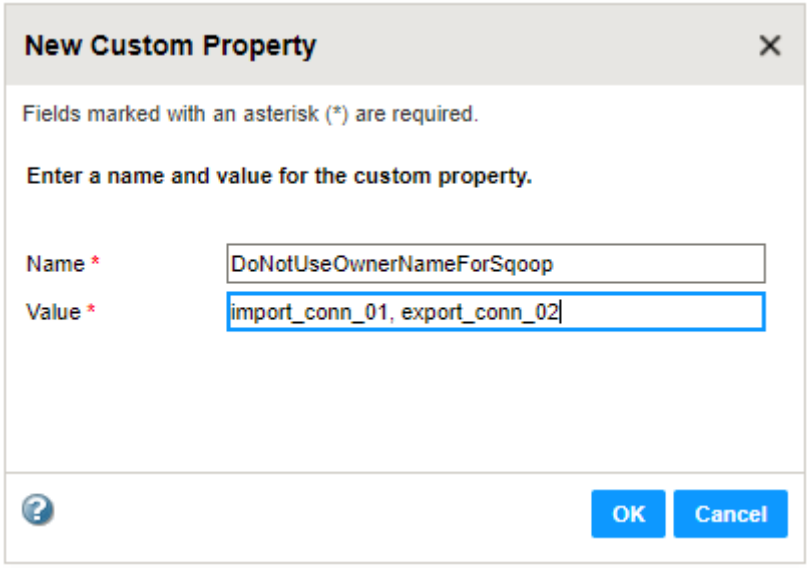
Use the Administrator tool to add the DoNotUseOwnerNameForSqoop custom property to the Enterprise Data Preparation Service. Enter the name of each JDBC connection that Enterprise Data Preparation uses to import and export data as the custom property value.

The DoNotUseOwnerNameForSqoop custom property enables Sqoop to use the connections to access a table in the Azure SQL Database target without requiring the name of the schema that contains the table. By default, Sqoop requires the name of the schema that contains the table to connect.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select the Enterprise Data Preparation Service.
3. Click the **Properties** view.
4. Click the pencil icon in the Custom Options section.
5. Click **New** in the Edit Custom Options dialog box.
6. Enter DoNotUseOwnerNameForSqoop as the custom property name.

7. Enter the name of each JDBC connection Enterprise Data Preparation uses to import and export data as the custom property value. If you enter multiple connection names, separate each connection name with a comma.

   The following image shows the **New Custom Property** dialog box:




8. Click **OK**.

You can now import and export data assets between Enterprise Data Preparation and Azure SQL Database.

For more information about importing and exporting data assets, see the Enterprise Data Preparation User Guide.

## Authors

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